Yash Kuletha

Data Engineer | Analytics Engineering

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SUMMARY

Data engineer with about 8 months of experience across two internships, building analytics-ready datasets with SQL and Python on lakehouse architectures. Delivered two Databricks projects (Medallion architecture, Delta Lake, Unity Catalog) with star-schema models, governed metric definitions, executive dashboards and audit-based data quality monitoring, plus validated ETL pipelines and Apache Iceberg tables on AWS at XCaliber Health. Focused on trusted metrics and self-service reporting that business stakeholders can rely on.

EXPERIENCE

Data Engineering / AI Infrastructure Intern — XCaliber Health *Bengaluru — Jan 2026 – June 2026*

- Built Python ETL pipelines for healthcare data ingestion and eCQM (electronic clinical quality measure) processing, with parsing and validation checks, cutting processing time by ∼50% and producing clean, analysis-ready data.
- Stored platform data as Apache Iceberg tables on AWS S3, registered in the AWS Glue Data Catalog, as the source for downstream processing; wrote tests ensuring every new tenant automatically gets its Iceberg table and catalog entry.
- Built distributed workflows with Ray and Daft to transform large-scale structured and unstructured healthcare data, improving throughput by ∼40% over single-node methods.
- Built Go + Temporal workflows with automated retries and duplicate-execution prevention so data pipelines ran without manual reruns on failed runs.
- Built an event-driven AWS pipeline in which tenant SOP PDFs stored in S3 trigger a Lambda function that starts a Temporal workflow to build a knowledge graph from the document.
- Developed backend REST APIs and internal SDK support so data capabilities could be reused across platform services, and wrote pytest suites for my codebases.

Automation and Analytics Intern — L&T Technology Services *Vadodara — May 2024 – July 2024*

- Cleaned and analyzed recruitment data in Excel, improving data quality and consistency for reporting.
- Built Power BI dashboards surfacing recruitment metrics and trends, replacing manual reporting with self-service views for stakeholders.
- Built a Resume Analyzer in Power Apps and Power Automate, automating candidate-screening steps.

PROJECTS

QuickSight Retail — Multi-Source Retail Analytics Lakehouse *Github*

Databricks, PySpark, SQL, Lakeflow Connect, Auto Loader, Delta Lake, Unity Catalog, Genie

- Integrated Salesforce CRM, PostgreSQL and transaction CSVs into a Bronze–Silver–Gold lakehouse using Lakeflow Connect, Auto Loader and Lakeflow Spark Declarative Pipelines, with a Gold star schema of fact and dimension tables for sales analysis.
- Defined a Unity Catalog metric view (6 business measures, 15+ dimensions) so sales KPIs are calculated consistently across teams.
- Built an executive dashboard and a Genie natural-language interface, letting business users explore sales performance without writing SQL.

NeoBank — Metadata-Driven Incremental Banking Pipeline *Github*

Databricks, PySpark, Spark SQL, Delta Lake, JDBC, Auto Loader, Unity Catalog

- Built a config-driven Bronze–Silver–Gold pipeline ingesting SQL Server and cloud-storage data via JDBC and Auto Loader, using metadata, watermark and audit tables to track incremental-load state and execution status per layer.
- Produced analytics-ready Gold datasets for Customer 360, branch performance, transaction channels, daily KPIs and customer risk summaries.
- Orchestrated all layers through a master Databricks job with automated email reports (per-table status, record counts, errors); added an executive dashboard and Genie agent.

TinyML on Microcontrollers — Published Research Paper *Github*

Python, Jupyter, TinyML, Autoencoders, Anomaly Detection, Arduino

- Implemented an autoencoder anomaly detection model on Arduino Nano BLE Sense Rev 2 using the ECG5000 dataset, optimized for on-device memory and compute limits.
- Co-authored "TinyML on Microcontrollers: A Review" (Journal of Advanced Research in Electronics Engineering and Technology, Vol. 13, Issue 1, 2026).

TECHNICAL SKILLS

- **Languages:** SQL, Python, Java, Go
- **SQL & Data Modeling:** Dimensional Modeling (Star Schema, Fact/Dimension Tables), Window Functions, Spark SQL, Joins and Aggregations, Metric Definitions, Metric Views
- **Lakehouse & Data Engineering:** Medallion Architecture (Bronze/Silver/Gold), Delta Lake, Apache Iceberg, ETL/ELT, Incremental Loading, PySpark, Apache Spark, Auto Loader, Lakeflow Spark Declarative Pipelines, JDBC

- **Platforms & Cloud:** Databricks (Jobs/Workflows, Dashboards, Genie), Unity Catalog, AWS (S3, Glue Data Catalog, Lambda, EC2, Secrets Manager), Azure SQL Database
- **Data Quality & Governance:** Data Validation, Audit and Metadata Tables, Run Monitoring and Status Reporting, Anomaly Detection, Unity Catalog Governance, Glue Data Catalog
- **BI & Visualization:** Power BI, Databricks Dashboards, Databricks Genie, Excel
- **Orchestration & Engineering Practices:** Temporal, Databricks Jobs, Ray, Daft, REST APIs, Git, GitHub Actions (CI/CD), pytest, Docker, Liquibase
- **Databases:** PostgreSQL, SQL Server, MySQL, Azure SQL, Cloud Object Storage
- **ML & Python Libraries:** Pandas, NumPy, Scikit-learn, TensorFlow, Autoencoders, TinyML

EDUCATION & CERTIFICATIONS

NIIT University — B.Tech, Computer Science (CGPA: 8.51) *Aug 2022 – Aug 2026*

- **Machine Learning Specialization** — Stanford University / DeepLearning.AI (Coursera), July 2025